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A genome-wide MAGIC kit for recombinase-independent mosaic analysis in *Drosophila*

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eLife Assessment

The study showcases a significant and **important** enhancement of the MAGIC transgenesis method, by extending it genome-wide to all chromosomes. The authors **convincingly** demonstrate that the MAGIC mosaic clones can be generated for genes from all, including the 4th chromosome. With this toolkit extension, the method is now most likely set to strongly rival the classical FRT/Flp recombination system for gene manipulation in flies.

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Abstract

Mosaic analysis has been instrumental in advancing developmental and cell biology. Most current mosaic techniques rely on exogenous site-specific recombination sequences that need to be introduced into the genome, limiting their application. Mosaic analysis by gRNA-induced crossing-over (MAGIC) was recently developed in *Drosophila* to eliminate this requirement by inducing somatic recombination through CRISPR/Cas9-generated DNA double-strand breaks. However, MAGIC has not been widely adopted because gRNA-markers, a required component for this technique, are not yet available for most chromosomes. Here, we present a complete, genome-wide gRNA-marker kit that incorporates optimized designs for enhanced clone induction and more effective clone labeling in both positive MAGIC (pMAGIC) and negative MAGIC (nMAGIC). With this kit, we demonstrate clonal analysis in a broad range of *Drosophila* tissues, including cell types that have been difficult to analyze using recombinase-based systems. Notably, MAGIC enables clonal analysis of pericentromeric genes and deficiency chromosomes and in interspecific hybrid animals, opening new avenues for gene function study, rapid gene discovery, and understanding cellular basis of speciation. This MAGIC kit complements existing systems and makes mosaic analysis accessible to address a wider range of biological questions.

Introduction

Mosaic animals containing genetically distinct populations of cells in the same organism are useful for *in vivo* studies of complex biological processes. For this reason, techniques that can generate genetically labeled mosaic clones have been utilized in both vertebrates and invertebrates to study tissue-specific functions of pleiotropic genes, developmental timing, cell lineages, cell proliferation, neural wiring, and many other biological phenomena [1–3]. The most popular mosaic techniques rely on site-specific recombination systems, such as FRT/Flp [4, 5] and LoxP/Cre [6, 7], to induce somatic recombination between homologous chromosomes. Such techniques require the introduction of recombination sites to specific locations in the genome and thus cannot be applied to unmodified chromosomes. To overcome this limitation, we recently developed a recombinase-independent mosaic technique called mosaic analysis by gRNA-induced crossing-over (MAGIC) [8]. In MAGIC, the CRISPR/Cas9 system generates double-strand breaks (DSBs) at a predefined genome location to induce homologous recombination in precursor cells during S/G2 phase. Subsequent chromosomal segregation during mitosis can result in clones homozygous for the chromosomal segments distal to the crossover site. Two variants of this technique in *Drosophila*, positive MAGIC (pMAGIC) and negative MAGIC (nMAGIC), label the resulting homozygous clones by the presence and absence of fluorescent markers, respectively [8]. Like FRT/Flp-based techniques, MAGIC enables characterization of homozygous clones of lethal mutations in otherwise heterozygous animals [8].

However, unlike FRT/Flp-based techniques, MAGIC does not require prior genetic modification of the test chromosome. Thus, it can potentially be used on any chromosome and have much wider applications. Foremost, mutations of diverse natures have been established for most *Drosophila* genes [9–14], and thousands of deficiency strains harbor deletions that collectively uncover 98.4% of the *Drosophila* genome [15]. However, most of these mutant chromosomes cannot be analyzed by traditional mosaic techniques due to the lack of FRT sites or incompatibility with the FRT/Flp system. In contrast, MAGIC can theoretically be applied to these chromosomes as they are, allowing convenient genome-wide mosaic screens. In addition, genes located more proximal to centromeres than existing FRT sites cannot be analyzed by FRT/Flp techniques. In comparison, MAGIC can potentially be used to study these genes because the crossover site in MAGIC can be flexibly defined by users. Lastly, given that MAGIC is compatible with wild-derived chromosomes [8], it may be able to generate homozygous clones of a chromosome derived from a single species in an interspecific hybrid animal, allowing the study of species-related cell-cell interactions.

Despite these potentials, MAGIC has not been widely adopted by the *Drosophila* community. A major barrier is the lack of gRNA-marker transgenes on most chromosomal arms, which are necessary for DSB induction and fluorescent clone labeling [8]. In addition, the existing gRNA-markers suffer from several limitations, including low frequency of clone induction, weak labeling of pMAGIC clones, and suboptimal visualization of nMAGIC clones. Because of these reasons, MAGIC has only been successfully applied to a few genes in a limited number of *Drosophila* tissues [8, 16].

To overcome these limitations, here we first optimized gRNA-marker designs to improve clonal induction, the brightness of pMAGIC clones, and visualization of nMAGIC clones. Then we generated pMAGIC and nMAGIC gRNA-markers for all chromosomal arms and characterized their ability to generate clones. Using this kit, we demonstrate mosaic analysis of centromere-proximal genes, deficiency chromosomes, chromosomes derived from different *Drosophila* species in interspecific hybrid animals. This kit allows optimal clone induction in diverse cell and tissue types and should be useful for studying a wide range of biological processes.

Results

New gRNA-marker designs improve pMAGIC and nMAGIC

MAGIC relies on a gRNA-marker dual-transgene (**Figure 1A**) inserted in a specific chromosomal arm to both induce and visualize clones homozygous for this arm [8]. The gRNA part of the construct expresses two gRNAs ubiquitously to target a pericentromeric location on this arm. Two gRNAs targeting two sequences that are close to each other, instead of a single gRNA, are used to increase the probability of DSBs and thus the clone frequency. The marker part in pMAGIC utilizes a ubiquitously expressed Gal80 to prevent Gal4-dependent labeling of heterozygous and homozygous cells for the gRNA-marker, while allowing labeling of homozygous cells that lose the gRNA-marker. In contrast, nMAGIC uses ubiquitously expressed BFP, which results in brighter labeling of gRNA-marker homozygous cells, intermediate labeling of gRNA-marker heterozygous cells, and lack of labeling of gRNA-marker-negative homozygous cells.

We have previously developed gRNA-marker vectors for both pMAGIC and nMAGIC. However, gRNA-markers made with these vectors exhibit some limitations. First, the clone frequency can be low for certain gRNAs [8]. Second, the *ubi-Gal80* in pMAGIC gRNA-markers does not completely suppress Gal4 activity in certain tissues, such as hemocytes. Third, pMAGIC clones are sometimes too dim to visualize cell morphology, such as thin dendrite and axon projections of neurons. Lastly, nMAGIC utilizes a nuclear BFP marker (nlsBFP), which does not show the cell shape and sometimes cannot mark clones effectively. Thus, to optimize MAGIC, we first sought to improve the gRNA-marker designs.

Since clone induction in MAGIC depends on gRNA-induced DNA DSBs, we asked whether a more efficient gRNA design enhances clone frequency in somatic tissues. The previous gRNA-marker vectors used a tgFE design, which contains the “F+E” gRNA scaffold and a tRNA^{Gly} before each gRNA targeting sequence (**Figure 1A**) [8]. However, we have shown that an improved Qtg2.1 design, which contains the gRNA2.1 scaffold and a single tRNA^{Gln} spacer between the two gRNAs (**Figure 1A**), is much more mutagenic than tgFE in somatic tissues [17]. We thus compared two pMAGIC gRNA-markers that are based on these two gRNA designs but target the same genomic sequences at cytological band 40D2 to assess their ability to induce clones in peripheral sensory neurons. When used with the same neuronal/epidermal precursor Cas9, *zk-Cas9* [8], Qtg2.1 resulted in three times more neuronal clones as compared to tgFE (**Figure 1B**), confirming a positive correlation between gRNA efficiency and clone frequency.

To ensure more complete suppression of Gal4 activity by Gal80 in pMAGIC, we replaced the *ubi* enhancer driving Gal80 expression with an *αTub84B* (*tub*) enhancer that was used in *tub-Gal80* in the MARCM system [18]. When combined with the larval hemocyte marker *pxn-Gal4 UAS-CD4-tdTom* [19] (**Figure 1C**), *ubi-Gal80* did not completely suppress the labeling of hemocytes (**Figure 1D**). In contrast, no labeled hemocytes could be detected in the presence of *tub-Gal80* (**Figure 1E**), suggesting that *tub-Gal80* is a better marker for pMAGIC.

To improve clone brightness in pMAGIC, we sought to destabilize Gal80 and reduce its expression, reasoning that dim clones are due to prolonged Gal80 activity after clone induction. In addition to replacing the *ubi* enhancer with the *tub* enhancer, we also introduced protein and mRNA destabilization sequences (DE) [20, 21] at the 3’ end of the Gal80 coding sequence, and replaced the *His2Av* polyA by *SV40* polyA (**Figure 1F**), as the latter reduces transgene expression [22]. By measuring the brightness of epidermal (**Figure 1G**) and neuronal (**Figure 1H**) clones induced by *gRNA-40D2* and *gRNA-42A4*, we found that each of the three changes improved clone brightness.

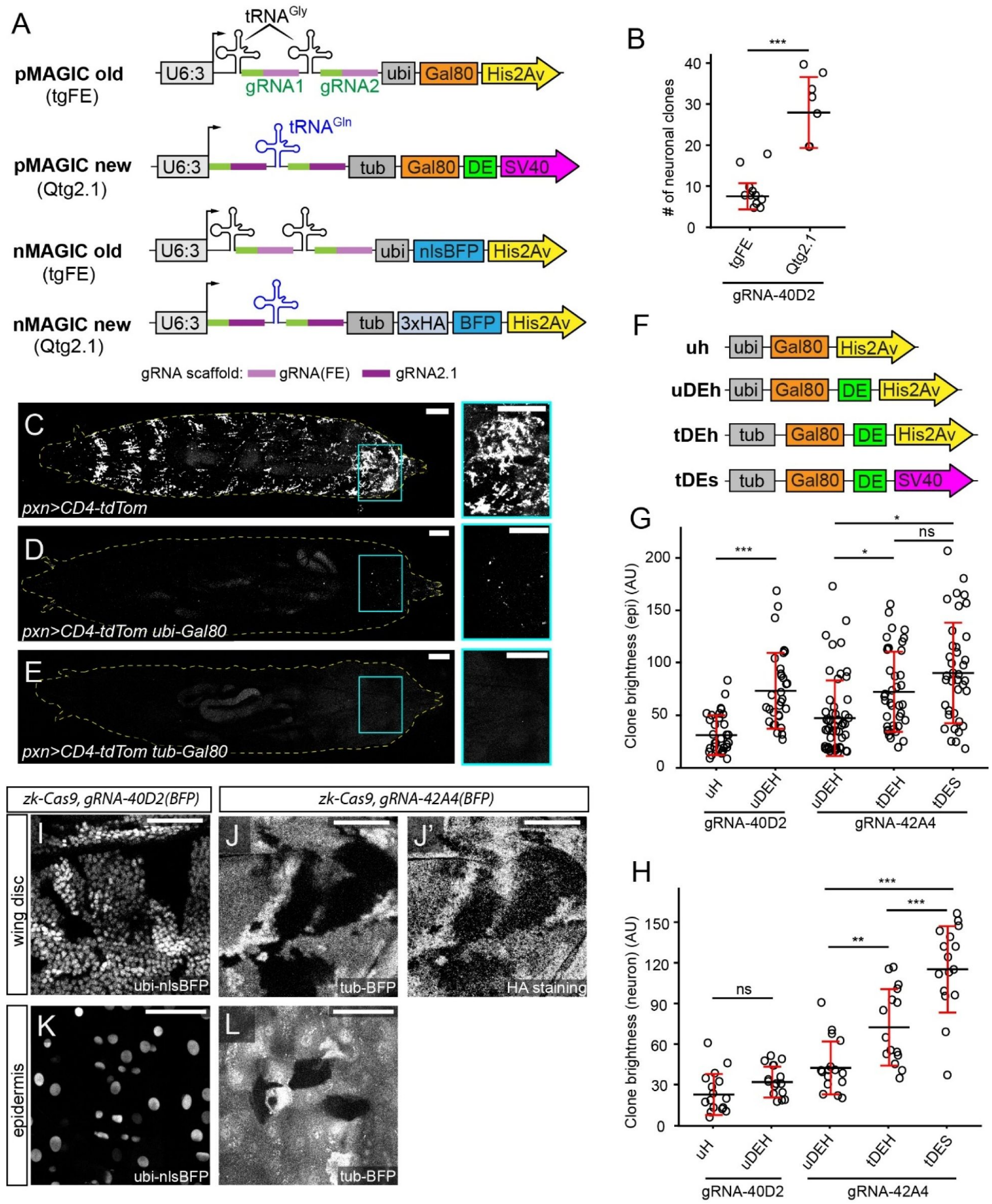

Figure 1.

New gRNA-marker designs improve pMAGIC and nMAGIC.

(A) Original and new designs of gRNA-markers for pMAGIC and nMAGIC. (B) Comparison of clone frequency in larval sensory neurons between two gRNA designs. The number represents clones between A1 and A7 segments on one side of each larva. n = larvae number: tgFE (n=10), Qtg2.1 (n=10). (C-E) Labeling of hemocytes in whole 3rd instar larvae by *pxn^{Gal4}>CD4-tdTom* alone (C) or together with *ubi-Gal80* (D) or *tub-Gal80* (E). The panels on the right show enlarged views of the boxed regions. (F) Designs of Gal80 variants tested in pMAGIC gRNA-markers. (G) The brightness of epidermal clones labeled by pMAGIC gRNA-markers. n = image numbers: gRNA-40D2-uH (n = 32), gRNA-40D2-uDEH (n = 31), gRNA-42A4-uDEH (n = 52), gRNA-42A4-tDEH (n = 39), gRNA-42A4-tDES (n = 38). (H) The brightness of neuronal clones labeled by pMAGIC gRNA-markers. n = neuron numbers: gRNA-40D2-uH (n = 16), gRNA-40D2-uDEH (n = 16), gRNA-42A4-uDEH (n = 16), gRNA-42A4-tDEH (n = 15), gRNA-42A4-tDES (n = 16). (I) Portion of a larval wing disc containing nMAGIC clones visualized by nlsBFP. (J and J') Portion of a wing disc containing nMAGIC clones labeled by cytosolic BFP (J) and HA staining (J'). (K) Epidermal clones on the larva body wall labeled by nlsBFP. (L) Epidermal clones visualized by cytosolic BFP. In all plots, black bar, mean; red bar, SD; AU, arbitrary unit. Student's t-test in (B); one-way analysis of variance (ANOVA) and Tukey's honest significant difference (HSD) test in (G) and (H). *p≤0.05, **p≤0.01, ***p≤0.001, ns, not significance. For (C-E), scale bar, 300 μm. For (I-M), scale bar, 100 μm.

Lastly, to visualize cell shape in nMAGIC, we replaced nlsBFP with cytosolic BFP tagged by 3X Hemagglutinin (HA), in addition to utilizing the *tub* enhancer. This new design allowed us to better discern clone shapes in both the wing imaginal disc and the epidermis (**Figure 1J, 1J**', and **1L**), in contrast to the previous *ubi-nlsBFP* design (**Figure 1I** and **1K**).

Thus, by altering the designs of the gRNAs and the Gal80 and BFP markers, we created new vectors optimized for more robust applications of nMAGIC and pMAGIC.

A gRNA-marker kit is established for all four chromosomes of *Drosophila*

To enhance the utility of MAGIC in *Drosophila*, we generated complete sets of pMAGIC and nMAGIC gRNA-markers for all four chromosomes (**Figure 2A**). To identify suitable gRNA target sites, we analyzed the pericentromeric sequences of X, 2L, 2R, 3L, 3R, and 4, based on three criteria [8]: conserved in closely related *Drosophila* species to minimize the chance of single nucleotide polymorphism, (2) located away from functionally critical regions to avoid disrupting essential processes, and (3) unique within the genome to minimize off-target effects. For each MAGIC construct, we selected a pair of non-repetitive gRNA target sequences in intergenic regions to maximize the chances of DSBs. These two sequences are closely linked to minimize the risk of large deletions. Given the variable efficiency of gRNA target sequences, we selected three pairs of gRNAs targeting three chromosomal locations for each chromosomal arm and named them according to the corresponding cytoband (**Table 1**).

To evaluate the clone-induction properties of these gRNAs, we combined the pMAGIC set with *zk-Cas9* and counted the number of neuronal clones in A1-A7 larval hemi-segments (**Figure 2B**). As expected, the clone frequency varied from gRNA to gRNA, but we were able to identify efficient gRNAs for every chromosomal arm. We previously found that different gRNAs follow the same trend of relative efficiency in different tissues [8]. Here we additionally tested nMAGIC gRNAs for 2R in the wing disc (**Figure 2C**) and noticed a similar trend of clone induction to that of their pMAGIC counterparts in sensory neurons (**Figure 2B**), suggesting that the results in sensory neurons are transferrable to other tissues.

Certain gRNAs (e.g. gRNA-40E1) exhibited very low clone frequency. Such gRNAs can be useful for inducing sparse clones in highly packed issues such as the brain. For example, using the same *ey-Cas9* [23], the highly efficient gRNA-40D2 induced too many neuronal clones in the adult brain for morphological analysis (**Figure 2D**), while gRNA-40E1 gave rise to few clones, whose projection patterns were much easier to analyze (**Figure 2E**).

Together, the pMAGIC and nMAGIC gRNA-marker lines constitute a complete kit (**Table 1**) for genome-wide MAGIC applications in *Drosophila*.

MAGIC allows clonal analysis in diverse tissues and cell types

To determine if MAGIC can be applied to diverse tissues in *Drosophila*, we conducted clonal analysis in the larva using *gRNA-40D2(Gal80)* and several tissue-specific Cas9s. With the ubiquitous *vas-cas9* [24] and *tub-Gal4 UAS-mCD8-GFP*, we readily detected clones in the larval brain (**Figure 3A**), proliferating tissues like eye and leg discs (**Figure 3B-C**), and polyploid tissues like the fat body, gut, and trachea (**Figure 3D-F**). Clone induction in polyploid tissues suggests that crossing-over events occurred before the last cell division. Using *zk-cas9* and *R38F11-Gal4>UAS-tdTomato (tdTom)*, we observed frequent epidermal clones (**Figure 3G**). Using the glia precursor *gcm-Cas9* and *repo-Gal4>UAS-mCD8-GFP*, we detected individual glial clones in the brain (**Figure 3H**). Lastly, the new pMAGIC gRNA-marker design allowed us to reliably induce hemocyte clones (**Figure 3I**). These results collectively demonstrate MAGIC's efficacy in generating clones in diverse *Drosophila* tissues, indicating its value in studying gene function and cell lineage in various tissue contexts.

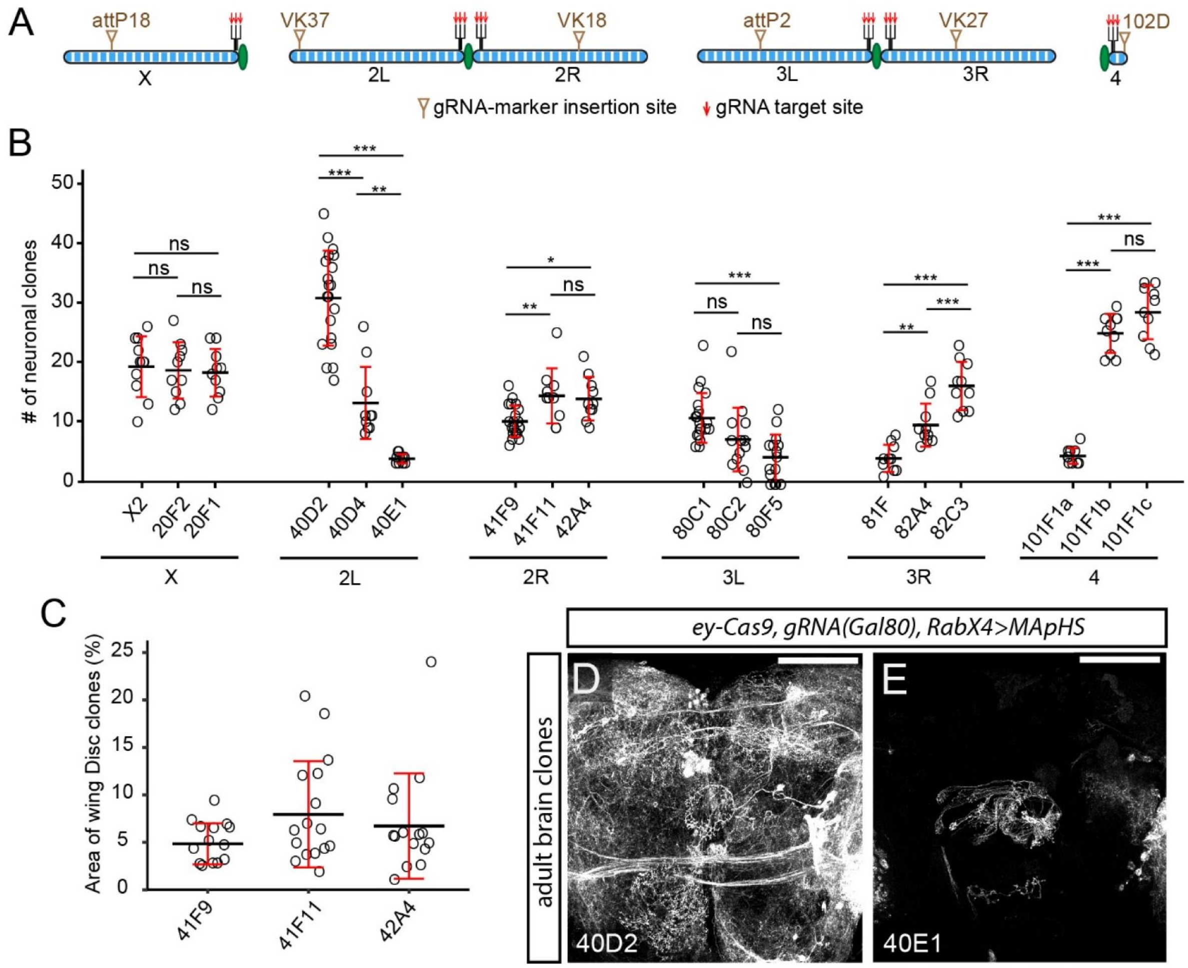

Figure 2.

A genome-wide gRNA-marker kit suits diverse needs of clone frequency.

(A) Scheme of gRNA-marker insertion sites and target sites on *Drosophila* chromosomes. (B) Comparison of clone frequencies of all pMAGIC gRNA-markers in larval sensory neurons, clones are labeled using *RabX4-Gal4 UAS-MApHs* (for Chromosome X, II and IV) or *21-7-Gal4 UAS-MApHs* (for Chromosome III). n = larvae number: X2 (n = 10), 20F2 (n = 10), 20F1(n = 10), 40D2 (n = 20), 40D4 (n = 10), 40E1 (n = 10), 41F9 (n = 20), 41F11 (n = 10), 42A4 (n = 10), 80C1 (n = 20), 80C2 (n = 14), 80F5 (n = 15), 81F (n = 10), 82A4 (n = 10), 82C3 (n = 10), 101F1a (n = 10), 101F1b (n = 10), 101F1c (n = 10). (C) Comparison of clone areas in larval wing discs labeled by nMAGIC gRNA-markers on 2R. n = wing disc number: 41F9 (n = 14), 41F11 (n = 16), 42A4 (n = 15). (D and E) Neuronal clones in the central part of the adult brain induced by pMAGIC gRNA-markers *gRNA-40D2* (D) and *gRNA-40E1* (E). In all plots, black bar, mean; red bar, SD. One-way ANOVA and Tukey's HSD test. *p≤0.05, **p≤0.01, ***p≤0.001, ns, not significance. For (D) and (E), scale bar 100 μm.

Chr Arm	target site	gRNA location	pMAGIC vector	nMAGIC vector	clone frequency
2L	40D2	attP[VK00037]	*pAC-U63-gRNA2.1-ubiGal80(DE)-His2Av*	*pAC-U63-tgRNA-nlsBFP**	31
2L	40D4	attP[VK00037]	*pAC-U63-tgRNA-Gal80**	*pAC-U63-tgRNA-nlsBFP**	13
2L	40E1	attP[VK00037]	*pAC-U63-tgRNA-Gal80**	*pAC-U63-tgRNA-nlsBFP**	4
2R	41F9	attP[VK00018]	*pAC-U63-gRNA2.1-ubiGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	10
2R	41F11	attP[VK00018]	*pAC-U63-gRNA2.1-ubiGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	14
2R	42A4	attP[VK00018]	*pAC-U63-gRNA2.1-tubGal80(DE)-SV40*	*pAC-U63-gRNA2.1-tubBFP(HA)*	14
3L	80F5	attP2	*pAC-U63-gRNA2.1-tubGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	11
3L	80C2	attP2	*pAC-U63-gRNA2.1-tubGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	7
3L	80C1	attP2	*pAC-U63-gRNA2.1-tubGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	4
3R	81F	attP[VK00027]	*pAC-U63-gRNA2.1-tubGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	4
3R	82A4	attP[VK00027]	*pAC-U63-gRNA2.1-tubGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	10

Table 1.

gRNA-marker collection

3R	82C3	attP[VK00027]	*pAC-U63-gRNA2.1-tubGal80(DE)-His2Av*	*pAC-U63-gRNA2.1-tubBFP(HA)*	16
X	X2	attP18	*pAC-U63-gRNA2.1-tubGal80(DE)-SV40*	*pAC-U63-gRNA2.1-tubBFP(HA)*	19
X	20F1	attP18	*pAC-U63-gRNA2.1-tubGal80(DE)-SV40*	*pAC-U63-gRNA2.1-tubBFP(HA)*	18
X	20F2	attP18	*pAC-U63-gRNA2.1-tubGal80(DE)-SV40*	*pAC-U63-gRNA2.1-tubBFP(HA)*	19
IV	101F1-a	attP[ZH-102D]**	*pAC-U63-gRNA2.1-tubGal80(DE)-SV40*	*pAC-U63-gRNA2.1-tubBFP(HA)*	4
IV	101F1-b	attP[ZH-102D]**	*pAC-U63-gRNA2.1-tubGal80(DE)-SV40*	*pAC-U63-gRNA2.1-tubBFP(HA)*	25
IV	101F1-c	attP[ZH-102D]**	*pAC-U63-gRNA2.1-tubGal80(DE)-SV40*	*pAC-U63-gRNA2.1-tubBFP(HA)*	28

Clone frequencies are measured by averaging clone numbers in 10 laterally mounted 3rd instar larvae with pMAGIC gRNA-markers. Only clones in A1 to A7 segments of one side of each larva were counted.

*Previously published gRNA-markers.

**3xP3-RFP has been removed from these lines.

Table 1. (continued)

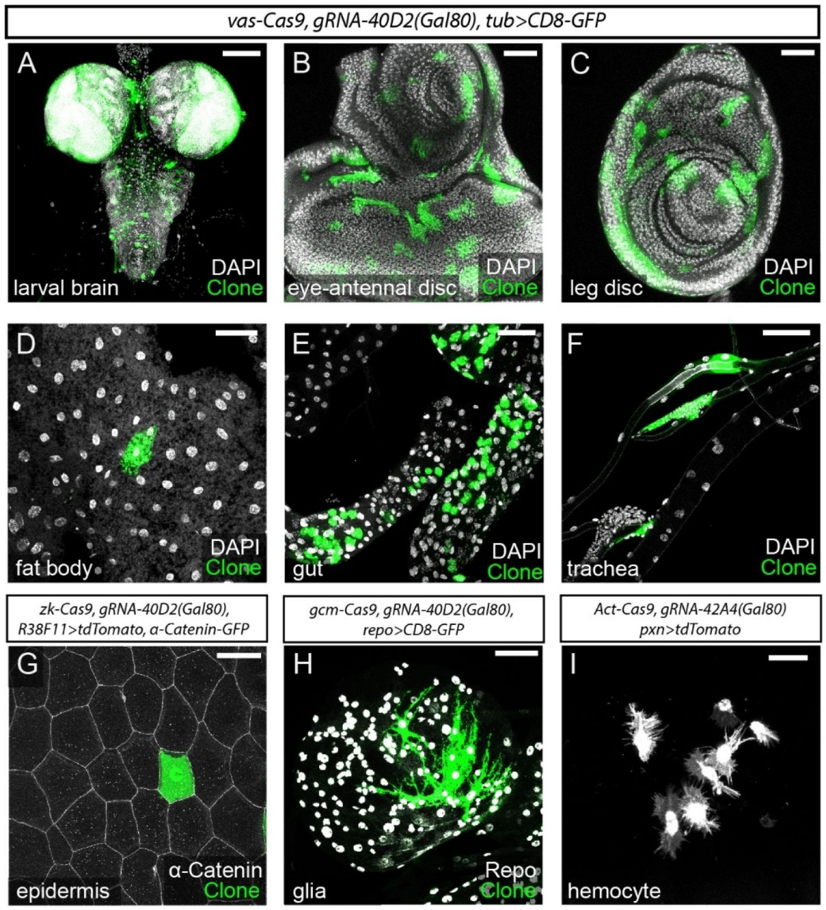

Figure 3.

MAGIC allows clonal analysis in diverse tissues and cell types.

(A-F) pMAGIC clones induced in different tissues by *vas-Cas9 gRNA-40D2(Gal80)* and labeled by *tub-Gal4 UAS-CD8-GFP* (green). DAPI staining (white) shows all nuclei. (G) A pMAGIC epidermal clone on the larval body wall induced by *zk-Cas9 gRNA-40D2(Gal80)* and labeled by *R38F11-Gal4 UAS-tdTom* (green). Epidermal junctions are labeled by *α-Catenin-GFP* (white). (H) pMAGIC glia clones in the larval brain induced by *gcm-Cas9 gRNA-40D2(Gal80)* and labeled by *repo-Gal4 UAS-CD8-GFP* (green). Glial nuclei are labeled by Repo staining (white). (I) pMAGIC hemocyte clones induced by *Act-Cas9 gRNA-40D2(Gal80)* and labeled by *pxn-Gal4 UAS-tdTom*. For figure A, D-F, H, scale bar 100 μm. For figure B-C, G, scale bar 50 μm. For Figure I, scale bar 25 μm.

The neuromuscular junction (NMJ) in *Drosophila* larvae has been a valuable model for elucidating synaptic biology, owing to its simplicity, accessibility, and conserved features shared with mammalian excitatory synapses [25–27]. Despite its popularity, the NMJ has rarely been analyzed by the MARCM technique in the literature, likely due to the difficulty of inducing clones in motor neurons. To test the effectiveness of MAGIC in analyzing gene function at the NMJ, we selected two genes crucial for synaptic function and construction: *Vesicular glutamate transporter* (*VGlut*) [28] and *bruchpilot* (*brp*) [29, 30], null mutations of each of which exhibit recessive lethality in larvae. Combined with appropriate gRNA(Gal80) lines (42A4 for *vGluT* on 2R and 40D2 for *brp* on 2L) and *tub-Gal4 UAS-CD8-GFP*, *zk-Cas9* induced frequent clones in type Ib boutons (**Figure 4A-B**”). The loss of vGluT or Brp specifically in GFP-labeled NMJs was confirmed by immunostaining. These results exemplify the value of pMAGIC for dissecting gene function in NMJ biology at the single cell level.

MAGIC enables clonal analysis of pericentromeric genes, 4th chromosome-associated mutations, and in interspecific hybrid animals

Because all FRT sites in the existing FRT/Flp mosaic system are located at some distances away from centromeres, it has not been possible to study genes located between the FRT sites and the corresponding centromeres by clonal analysis. In contrast, the gRNA target site in a MAGIC experiment can be user-selected to enable clonal analysis of any given gene. To illustrate this potential, we used *gRNA-41F9* to induce homozygous mutant clones of *Ecdysone receptor* (*EcR*), which is located at 42A10 and is inaccessible to all FRT sites on 2R. EcR is a transcription factor required for neuronal remodeling during metamorphosis [31]. In peripheral sensory neurons, EcR activation at the beginning of pupariation causes apoptosis of some dendritic arborization (da) neurons while triggering dendrite pruning of other da neurons [32]. As expected, a wildtype (WT) pMAGIC clone of class IV da (C4da) neuron exhibited complete dendrite pruning at 16 h after puparium formation (APF), accompanied by dendrite debris phagocytosed into epidermal cells [19] (**Figure 5A**). In contrast, *EcR* mutant da neuron clones still maintained larval dendritic arbors at 16 h APF (**Figures 5B-D**), instead of dying (in the case of C3da) or undergoing dendrite pruning (in the cases of C1da and C4da).

Mosaic analysis of genes located on the 4th chromosome had not been possible until recent introduction of FRT sites onto this chromosome by CRISPR-mediated knock-in [33]. Despite these advances, existing mutations on FRT-lacking 4th chromosomes still cannot be analyzed by the FRT/Flp system. A valuable gene-disruption resource in *Drosophila* is the deficiency library consisting of strains harboring chromosomal deletions. MAGIC can potentially be used in conjunction with the deficiency library to screen for genes important in a particular biological process. To test the potential of MAGIC for analyzing mutations on the 4th chromosome and for gene discovery with deficiencies, we generated pMAGIC C4da clones of *Df(4)ED6380*, a deficiency that deletes a segment between cytological bands 102B7 and 102D5. These clones show dramatically reduced dendrites compared to the WT controls (**Figure 4E**-**4G**), indicating the existence of genes important for dendrite growth in this region.

When examining nMAGIC gRNA-markers for the 4th chromosome, we noticed uneven expression of *tub-3xHA-BFP* in epidermal cells and some imaginal tissues, exemplified by cells lacking detectable BFP signal (**Figures S1A,S1B**). This variegated expression is likely due to transgenes residing in heterochromatin and is common for transgenes located on the 4th chromosome [34]. While this uneven expression limits the usefulness of our gRNA-markers in the corresponding epithelial tissues, Gal80 in pMAGIC gRNA-markers efficiently suppressed Gal4 activities in most neurons (**Figure S1C**), conforming their effectiveness in neuronal MAGIC analysis.

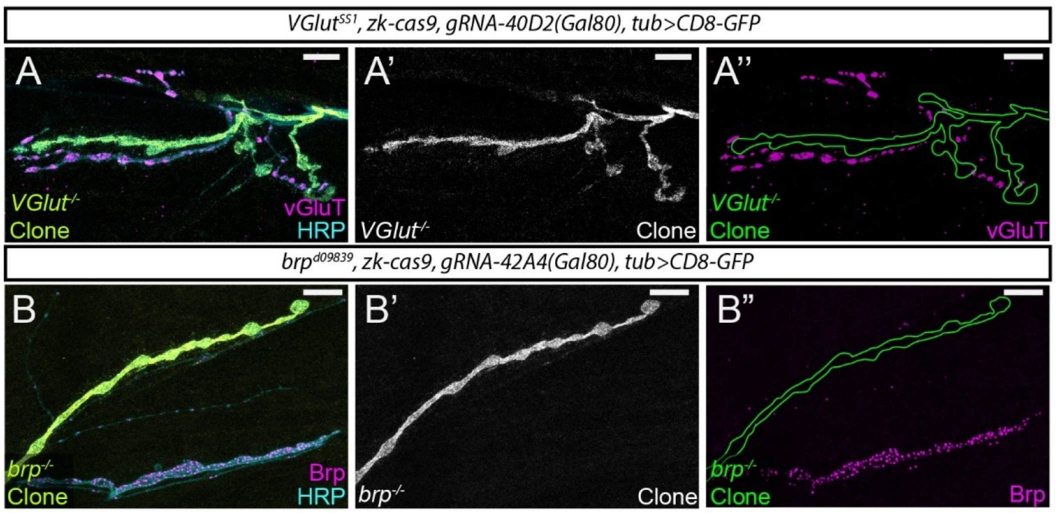

Figure 4.

MAGIC facilitates clonal analysis at the NMJ.

(A-A”) pMAGIC clones of *VGlut*^{1} mutation in motor neurons at the neuromuscular junction. Clones were induced by *zk-Cas9 gRNA-40D2(Gal80)* and labeled by *tub-Gal4 UAS-CD8-GFP*., The loss of VGlut is confirmed by VGlut staining. The mutant clones are outlined in (A”). (B-B”) A pMAGIC clone of *brp*^{d09839} mutation in a motor neuron at the neuromuscular junction. Clones were induced by *zk-Cas9 gRNA-42A4(Gal80)* and labeled by *tub-Gal4 UAS-CD8-GFP*. The loss of Brp is confirmed by Brp staining. The mutant clone is outlined in (B”). In both experiments, HRP staining shows all axons. Scale bars, 10 µm.

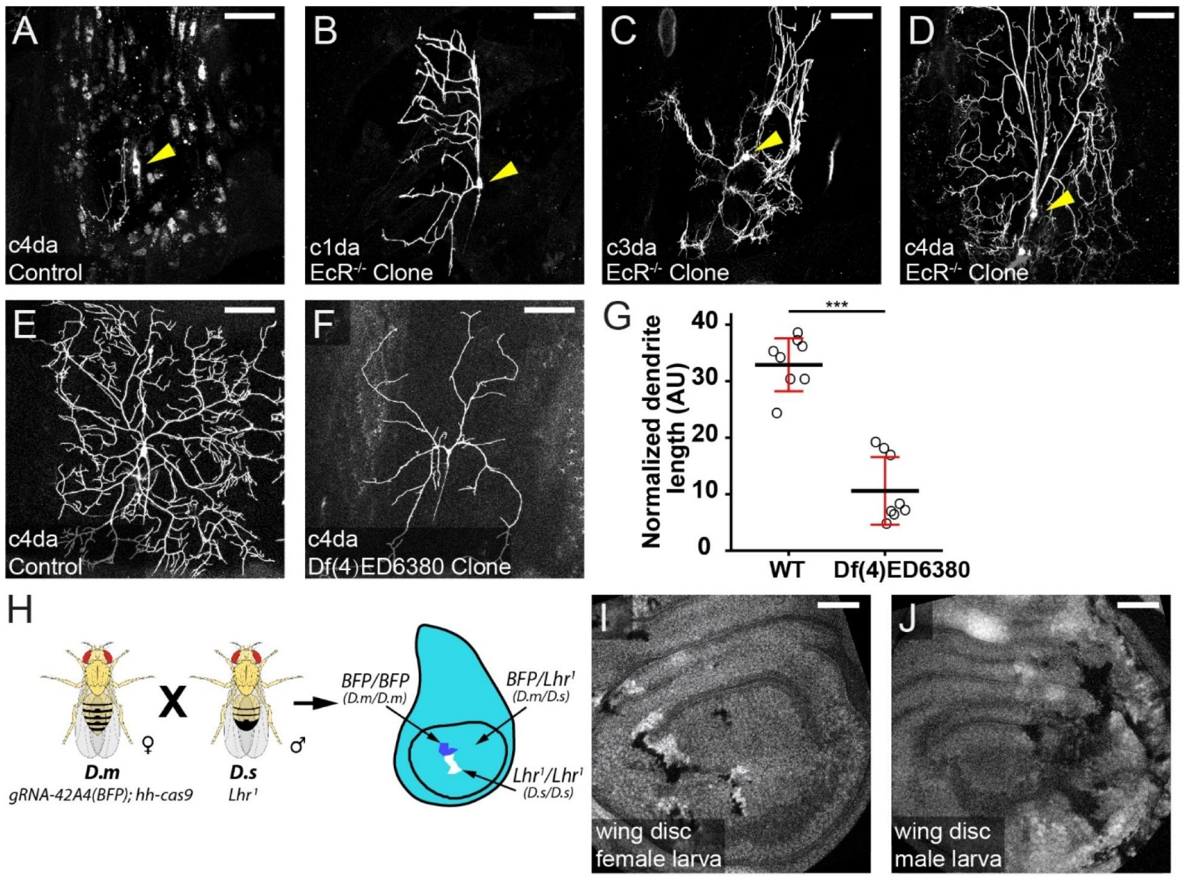

Figure 5.

MAGIC enables clonal analysis of pericentromeric genes, 4^{th} chromosome-associated mutations, and in interspecific hybrid animals.

(A) A WT pMAGIC class IV da neuron clone exhibiting complete dendrite pruning at 16 hours APF. (B-D) pMAGIC clones of *EcR*^{M554fs} mutation in da neurons imaged at 16 hours APF, exhibiting the lack of pruning (B and D) or apoptosis (C). In (A-D), the clones were induced by *zk-cas9* with *gRNA-41F9(Gal80)* and labeled by *RabX4-Gal4 UAS-MApHS*. Neuronal cell bodies are indicated by arrows. MApHS contains pHluorin and tdTom [19], but only tdTom signals are shown. The signals in epidermal cells (A) were due to engulfment of pruned dendrites by epidermal cells [19]. (E and F) WT (E) and *Df(4)ED6380* (F) pMAGIC clones in C4da neurons induced by *zk-cas9 gRNA-101Fc(Gal80)* and labeled by *RabX4-Gal4 UAS-MApHS*. Only tdTom signals are shown. (G) Normalized dendrite length of WT clones and deficiency clones. Black bar, mean; red bar, SD. Student's t-test. ***p≤0.001. (H) Scheme for interspecific crosses between *D. melanogaster* (*D.m*) and *D. simulans* (*D.s*). (I and J) Wing discs from male (I) and female (J) progeny carrying clones. Scale bars, 50 μm.

Lastly, we wondered if the MAGIC system can generate clones in interspecific hybrid animals derived from *D. melanogaster* and *D. simulans* parents, given that the two species show a large degree of synteny [35]. The clones in these animals would contain homozygous chromosomal arms derived from a single species. To test this idea, we crossed *D. melanogaster* females containing *gRNA-42A4(BFP); hh-cas9* to *D. simulans* males carrying a loss of function mutation of *Lethal hybrid rescue mutation* (*Lhr*), which ensures the viability of the hybrid male progeny [36]. Indeed, we observed twin spots consisting of dark clones, containing only *D. simulans* 2R, and brighter clones, containing only *D. melanogaster* 2R, in wing discs of both female and male progeny (**Figure 5G** and **5H**), demonstrating the feasibility of studying species-specific alleles in cell-cell interactions in interspecific hybrids.

Discussion

Conceptually, MAGIC is a simpler and more convenient mosaic technique compared to traditional recombinase-dependent methods. However, the lack of gRNA-marker transgenes has prevented its wide application in *Drosophila*. In this study, we present a complete gRNA-marker kit that enables genome-wide MAGIC in *Drosophila*, removing this bottleneck. The new gRNA-markers incorporate optimized designs that improve clone frequency and labeling in both pMAGIC and nMAGIC. We further demonstrate the compatibility of MAGIC with broad tissues and cell types. More importantly, MAGIC enables mosaic analyses that could not be accomplished by existing FRT/Flp systems, such as those of pericentromeric genes, deficiency chromosomes, and interspecific homologous chromosomes. Thus, this MAGIC kit provides *Drosophila* researchers with greater flexibility for conducting mosaic analyses of diverse purposes.

To implement MAGIC, one needs to first choose a proper Cas9 that is expressed in the precursor cells of the targeted cell population. We show that ubiquitously expressed Cas9 lines, such as *vas-Cas9* [24] and *Act5C-Cas9* [37], are sufficient to induce clones in broad tissues. Alternatively, heat shock (HS)-induced Cas9 [38] can offer temporal control of clone induction in most tissues, akin to the HS-Flp commonly used in FRT-based mosaic analysis. The third option is a Cas9 driven by a tissue-specific enhancer specifically in the precursor cells of the target tissues. Examples shown in this study include *ey-Cas9* expressed in many neuronal lineages [23], *gcm-Cas9* expressed in glial progenitor cells [39], *ZK-Cas9* expressed in precursor cells of epidermal cells, motor neurons, and somatosensory neurons [17], and *hh-Cas9* expressed in imaginal tissues [40]. These Cas9 lines have the advantages of being more specific and efficient, and more convenient to use than ubiquitous or inducible ones. Multiple strategies, including enhancer fusion [37, 40], Gal4-to-Cas9 conversion [17, 41], and insertion of Cas9 in specific gene loci [39], have been developed to ease the generation of such tissue-specific Cas9 lines. As an ongoing effort, we have been converting Gal4 lines known to be expressed in progenitor cells into Cas9 (available at *https://bdsc.indiana.edu/stocks/genome_editing/crispr_cas9.html*), in the hope of making MAGIC accessible to more *Drosophila* tissues.

The second component of MAGIC is a gRNA-marker line that resides on the appropriate chromosome arm and targets Cas9 to cut a pericentromeric site on the same arm. In the gRNA-marker kit, we generated three lines targeting different sites for each chromosome arm. These lines exhibit a broad range of clone frequencies in larval sensory neurons. As different gRNA-markers maintain similar relative efficiencies of clone induction across tissues [8], one can choose a gRNA-marker more appropriate for their applications based on the target gene location and the desired clone frequency. Notably, a higher clone frequency may not always be beneficial, such as when studying the projection patterns of individual neurons in the brain. A low efficiency gRNA-marker in this kit may be more desirable in such applications.

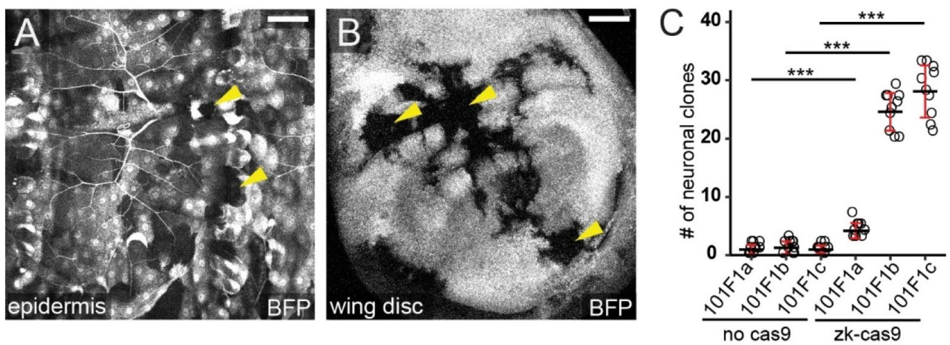

Figure S1.

Transgenic markers on the 4th chromosome show uneven expression.

(A-B) Representative epidermal (A) and wing disc (B) images showing uneven expression of *gRNA-101F1c(BFP)* inserted at *attP*^{102D} on the 4th chromosome. Yellow arrowheads point to cells lacking BFP expression. Scale bars, 50 µm. (C) Frequency of labeled neurons by indicated gRNA(Gal80) in the absence and presence of Cas9. The *zk-Cas9* dataset is the same as that for chromosome 4 in **Figure 2B**. Black bar, mean; red bar, SD. One-way ANOVA and HSD test. ***p≤0.001.

Besides commonly conducted mosaic analysis in *Drosophila*, MAGIC also enables many novel analyses that are difficult or impossible to accomplish with traditional systems. One example is to analyze interactions among species-specific cells in an interspecific hybrid animal for understanding the cell biological basis of hybrid incompatibility. Interspecific crosses between WT *D. melanogaster* and WT *D. simulans* result in sex-specific lethality of F1 progeny [36]. The cellular basis of this lethality is poorly understood. With MAGIC, one may generate mixed cell populations that are homozygous or heterozygous for species-specific alleles in the same hybrid embryo or larva, in which the relative fitness of the three cell populations can be compared. Similarly, MAGIC could complement genome-wide association studies (GWAS) using wild-derived isogenic strains and provide much more mechanistic detail. Such strains as those in the *Drosophila* Genetic Reference Panel (DGRP) [42] have been very useful for discovering loci that are responsible for phenotypic variations. MAGIC can be combined with these strains to analyze the effects of homozygosity of specific variants at the cell biology level [8]. In addition, MAGIC can be combined with the *Drosophila* deficiency kit for rapid genome-wide mosaic screens. A deficiency exhibiting a desired phenotype in such screens can be further dissected with smaller deficiencies within the deleted region or existing mutations in candidate genes. In this way, the convenience of MAGIC could significantly accelerate phenotype-based gene discovery. Lastly, because the crossover site in MAGIC can be user-defined, one can easily generate gRNA-markers that induce somatic recombination at specific genome locations. Such flexibility could be useful for mapping undefined genetic loci that are responsible for certain phenotypes, especially in non-traditional model organisms.

Materials and methods

Fly Stocks and Husbandry

See the Key Resource Table (Table S1) for details of fly stocks used in this study. Most fly lines were either generated in the Han lab or obtained from the Bloomington *Drosophila* Stock Center. *lethal hybrid rescue* (*lhr*) mutant *D. Simulans* was a gift from Dr. Dan Barbash. All flies were grown on standard yeast-glucose medium, in a 12:12 light/dark cycle, at 25°C unless otherwise noted. Virgin males and females for mating experiments were aged for 3-5 days.

To generate and label pMAGIC clones in larval peripheral sensory neurons, we used either *RabX4-Gal4 UAS-MApHS* (for gRNA-markers on chromosomes X, II, and IV) or *21-7-Gal4 UAS-MApHS* (for gRNA-markers on chromosome III) combined with *zk-cas9*. To count peripheral sensory neuronal clones on the larval body wall, third instar larvae were mounted laterally on slides and then counted in segment A1-A7 under a Nikon SMZ18 stereomicroscope. pMAGIC clones were induced and labeled by *RabX4-Gal4 UAS-MApHS* combined with *ey-cas9* in the fly adult brain, by *tub-Gal4 UAS-mCD8-GFP* combined with *vas-cas9* in larval brains, imaginal discs, fat bodies, guts, and trachea, by *repo-Gal4 UAS-mCD8-GFP* combined with *gcm-cas9* in glia, by *pxn-Gal4 UAS-tdTom* combined with *Act-cas9* in hemocytes, by *tub-Gal4 UAS-mCD8-GFP* combined with *zk-cas9* in larval motor neurons, by *R38F11-Gal4 UAS-tdTom* combined with *zk-cas9* in the larval epidermis. To induce nMAGIC clones in wing imaginal discs of interspecific hybrid animals, we use *gRNA-42A4(BFP)*; *hh-cas9* virgin females of *D. melanogaster* to cross with *Lhr*^{1} [43] *D. simulans* males.

Molecular Cloning

MAGIC gRNA cloning vectors: pMAGIC gRNA-marker vectors constructed in this study include pAC-U63-gRNA2.1-ubiGal80(DE)-His2Av, pAC-U63-gRNA2.1-tubGal80(DE)-His2Av and pAC-U63-gRNA2.1-tubGal80(DE)-SV40. To make pAC-U63-gRNA2.1-ubiGal80(DE)-His2Av, a fragment containing gRNA2.1 scaffold (gRNA2.1) and U6 3' flanking sequence (U63fl) was first used to replace gRNA2.1-QtRNA-Gal80(TS)-gRNA2.1-U63fl in pAC-U63-QtgRNA2.1-8R (Addgene 170514) [17]. Then a destabilization sequence (DE) containing a GS linker, amino acids (AAs) 422-461 of mouse ornithine decarboxylase [21], 2X RNA destabilizing nonamer (TTATTTATTgatccTTATTTATT)

[21] was added to the C-terminus of Gal80 in frame. To make pAC-U63-gRNA2.1-tubGal80(DE)-His2Av, a 2.6 kb *tub* enhancer was amplified from pENTR221-tubP [16] by PCR and used to replace the *ubi* enhancer in pAC-U63-gRNA2.1-ubiGal80(DE)-His2Av. To make pAC-U63-gRNA2.1-tubGal80(DE)-SV40, a SV40 polyA sequence was amplified from pAPIC-PHCS [22] and used to replace the His2Av polyA in pAC-U63-gRNA2.1-ubiGal80(DE)-His2Av. To make the nMAGIC gRNA-marker vector pAC-U63-gRNA2.1-tubBFP(HA), the gRNA2.1-U63fl fragment was first used to replace gRNA2.1-QtRNA-BFP(TS)-gRNA2.1-U63fl in pAC-U63-QtgRNA2.1-BR (Addgene 170513) [17]. The *tub* enhancer was then used to replace the *ubi* enhancer. Lastly, a synthetic DNA fragment (GenScript) encoding 3X HA was used to replace the nuclear localization signal at the N-terminus of mTagBFP. Cloning was carried out by ligation with T4 ligase or NEBuilder DNA Assembly reactions (New England Biolabs Inc.).

MAGIC gRNA expression vectors: 33 gRNA expression vectors were constructed with the corresponding gRNA cloning vectors as listed in **Table 1** according to published protocols [17]. Briefly, for each expression vector, two primers containing appropriate gRNA target sequences (Table S2) were used to amplify a fragment consisting of 3' end of U6:3 promotor, the first target sequence (TS1), gRNA2.1, tRNA^{Q}, the second target sequence (TS2), and the beginning sequence of gRNA2.1 with pAC-U63-QtgRNA2.1-BR as the PCR template. The PCR product was then assembled with SapI-digested gRNA cloning vectors using NEBuilder DNA Assembly.

Injections were carried out by Rainbow Transgenic Flies (Camarillo, CA 93012 USA) or Genetivision (Stafford, TX 77477) to transform flies through ϕC31 integrase-mediated integration into attP docker sites. The 3xP3-RFP selection marker in gRNA-markers inserted at the attP[ZH-102D] site were removed by crossing to Cre.

Live Imaging

Live imaging of larval epidermal cells, sensory neurons, and hemocytes was performed as previously described [44]. Animals were collected at 96 (for late third larvae) or 120 hours AEL (for wandering third instar larvae) hours after egg-laying (AEL) and mounted in glycerol on a slide with vacuum grease as a spacer. Animals were imaged using a Leica SP8 confocal microscope with a 40X NA1.3 oil objective, pinhole size 2 airy units, and a z-step size of 1 µm. For epidermis, images were taken at the dorsal midline of A2 and A3 segments. For dendritic arborization neurons, images were taken from A1 to A7 hemi-segments.

For imaging sensory neurons in pupae, newly formed pupae were collected and incubated at 25 °C. After 16 hours, the pupal cases were carefully removed, and the pupae were mounted dorsal side up on slides with halocarbon oil beneath a coverslip. Double-sided tape was used as a spacer. The mounted pupae were imaged using a Leica SP8 confocal microscope with a 40X NA1.3 oil objective.

Imaging of wing discs and other larval tissues

Larval dissections were performed as described previously [40]. Briefly, wandering third instar larvae were dissected in a small petri dish filled with cold phosphate-buffered saline (PBS). The anterior half of the larva was inverted. To prepare imaginal discs, trachea and gut were removed. Samples were then transferred to 4% formaldehyde in PBS and fixed for 20 minutes at room temperature. After washing with PBS, the tissues were stained in DAPI (1:1000) in 0.2% PBST (PBS with 0.2% Triton X-100) for 5 minutes. The tissues were washed again in PBST and mounted in SlowFade Diamond Antifade Mountant (Thermo Fisher Scientific) on a glass slide.

A coverslip was lightly pressed on top. Imaginal discs were imaged using a Leica SP8 confocal microscope with a 20X NA0.8 oil objective.

Adult brain imaging

Flies were aged for 1 day after eclosion. Brains were dissected in PBS at room temperature and then fixed in 4% paraformaldehyde in PBS with constant circular rotation for 20 minutes at room temperature. The brains were subsequently washed in 0.2% PBST and mounted on a glass slide under a glass coverslip. Vacuum grease was used as a spacer between the coverslip and the slide. Brains were imaged using a Leica SP8 confocal microscope with a 40X NA1.3 oil objective.

Larval fillet preparation

Larval fillet dissection was performed on a petri dish half-filled with PMDS gel. Wandering third instar larvae were pinned on the dish in PBS dorsal-side up and then dissected to expand the body wall. PBS was then removed and 4% formaldehyde in PBS was added to fix larvae for 15 minutes at room temperature. For VGlut staining, the fillets were fixed in Bouin's solution for 5 minutes at room temperature. Fillets were rinsed and then washed at room temperature in PBS for 20 minutes or until the yellow color from Bouin's solution faded. After immunostaining, the head and tail of fillets were removed, and the remaining fillets were placed in SlowFade Diamond Antifade Mountant on a glass slide. A coverslip was lightly pressed on top. Larval fillets were imaged using a Leica SP8 confocal microscope with a 40X NA1.3 oil objective.

Immunohistochemistry

Larval brains and larval fillets were rinsed and washed at room temperature in 0.2% PBST after fixation. The samples were then blocked in PBST with 5% normal donkey serum (NDS) for 1 hour before incubating with appropriate primary antibodies in the blocking solution. Brains were stained for 2 hours at room temperature, and fillets were stained overnight at 4°C. After additional rinsing and washing, the samples were incubated with secondary antibodies for 2 hours at room temperature. The samples were then rinsed and washed again before mounting and imaging. Primary antibodies used in this study are mouse anti-Repo antibody 8D12 (1:50 dilution), mouse anti-Brp antibody nc82 (1:100 dilution), rabbit anti-VGlut (1:200 dilution) [39], mouse anti-HA antibody (12CA5, 1:100), and goat anti-HRP conjugated with Cy3 (1:200). Secondary antibodies include donkey anti-mouse antibody conjugated with Cy5 (1:400) and donkey anti-rabbit antibody conjugated with Cy5 (1:400).

Image Analysis and Quantification

ImageJ analyses were conducted in Fiji/ImageJ. To compare brightness of clones in sensory neurons and epidermal cells, clones were detected based on thresholds to generate masks. The clone brightness within the masks was then measured as mean pixel intensity. To quantify clones in wing discs, 2-3 slices of optical sections in the middle of the disc were projected into a 2-dimensional (2D) image. Clones were detected based on a fixed threshold to generate masks, and the total area of clones in the wing pouch of each disc was measured.

The tracing and measurement of da neuron dendrites were done as previously described in detail [44]. Briefly, dendrites were segmented using local thresholding. The segments were then converted into single-pixel-width skeletons. The total length of skeletons was calculated based on pixel distance. Normalized dendrite length was calculated as dendritic length (μm)/segment width (μm).

Statistical Analysis

One-way analysis of variance (ANOVA) with Tukey's honest significant difference (HSD) test was used when dependent variable was normally distributed and there was approximately equal variance across groups. Student's t-test was used when two groups are compared. For additional information on the number of samples, see figure legends. R studio was used for all statistical analyses.

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Additional files

Table S1

Table S2

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Reviewer #1 (Public review):

Summary:

In this manuscript, Shen et al. have improved upon the mitotic clone analysis tool MAGIC that their lab previously developed. MAGIC uses CRISPR/Cas9-mediated double-stranded breaks to induce mitotic recombination. The authors have replaced the sgRNA scaffold with a more effective scaffold to increase clone frequency. They also introduced modifications to positive and negative clonal markers to improve signal-to-noise and mark the cytoplasm of the cells instead of the nuclei. The changes result in increase in clonal frequencies and marker brightness. The authors also generated the MAGIC transgenics to target all chromosome arms and tested the clone induction efficacy.

Strengths:

MAGIC is a mitotic clone generation tool that works without prior recombination to special chromosomes (e.g., FRT). It can also generate mutant clones for genes for which the existing FRT lines could not be used (e.g., the genes that are between the FRT transgene and the centromere).

This manuscript does a thorough job in describing the method and provides compelling data that support improvement over the existing method.

Weaknesses:

It would be beneficial to have a greater variety of clonal markers for nMAGIC. Currently, the only marker is BFP, which may clash with other genetic tools (e.g., some FRET probes) depending on the application. It would be nice to have far-red clonal markers.

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Reviewer #2 (Public review):

Summary:

In this study, the authors present the latest improvement of their previously published methods, pMAGIC and nMAGIC, which can be used to engineer mosaic gene expression in wild-type animals and in a tissue-specific manner. They address the main limitation of MAGIC, the lack of gRNA-marker transgenes, which has hampered the broader adoption of MAGIC in the fly community. To do so, they create an entire toolkit of gRNA markers for every Drosophila chromosome and test them across a range of different tissues and in the context of making Drosophila species hybrid mosaic animals. The study provides a significant and broadly useful improvement compared to earlier versions, as it broadens the use-cases for transgenic manipulation with MAGIC to virtually any subfield of Drosophila cell biology.

Strengths:

Major improvements to MAGIC were made in terms of clone induction efficiency and usability across the Drosophila model system, including wild-type genotypes and the use in non-melanogaster species.

Notably, mosaic mutants can now be created for genes residing on the 4th chromosome, which is exciting and possibly long-awaited by 4th chromosome gene enthusiasts.

Selection of the standard set of gRNA markers was done thoughtfully, using non-repetitive conserved and unique sequences.

The authors demonstrate that MAGIC can be used easily in the context of interspecific hybrids. I believe this is a great advancement for the Drosophila community, especially for evolutionary biologists, because this may allow for easy access to mechanistic, tissue-specific insight into the process of a range of hybrid incompatibilities, an important speciation process that is normally difficult to study at the level of molecular and cell biology.

In the same way, because it is not limited to usage in any particular genetic background, genome-wide MAGIC can be potentially used in wild-type genotypes relatively easily. This is exciting, especially because natural genetic diversity is rarely investigated more mechanistically and at the scale/resolution of cells or specific tissues. Now, one can ask how a particular naturally occurring allele influences cell physiology compared to another (control) while keeping the global physiological context of the particular genetic background largely intact.

Weaknesses:

It is not entirely clear how functionally non-critical regions were evaluated, besides that they are selected based on conservation of sequence between species. It may be useful to directly test the difference in viability or other functionally relevant phenotype for flies carrying different markers. Similarly, the frequency of off-targets could be investigated or documented in a bit more detail, especially if one of the major use-cases is meant for naturally derived, diverse genetic backgrounds. It is, at the moment, unclear how consistently the clones are induced for each new gRNA marker across different WT genetic backgrounds, for example, a set of DGRP genotypes, which could be highly useful information for future users.

https://doi.org/10.7554/eLife.108453.1.sa2

Reviewer #3 (Public review):

Summary:

In the manuscript by Shen, Yeung, and colleagues, the authors generate an improved and expanded Mosaic analysis by gRNA-induced crossing-over (MAGIC) toolkit for use in making mosaic clones in Drosophila. This is a clever method by which mitotic clones can be induced in dividing cells by using CRISPR/Cas9 to generate double-strand breaks at specific locations that induce crossing over at those locations. This is conceptually similar to previous mosaic methods in flies that utilized FRT sites that had been inserted near centromeres along with heat-shock inducible FLPase. The advantage of the MAGIC system is that it can be used along with chromosomes lacking FRT sites already introduced, such as those found in many deficiency collections or in EMS mutant lines. It may also be simpler to implement than FRT-based mosaic systems. There are two flavors of the MAGIC system: nMAGIC and pMAGIC. In nMAGIC, the main constituents are a transgene insertion that contains gRNAs that target DNA near the centromere, along with a fluorescent marker. In pMAGIC, the main constituents are a transgenic insertion that contains gRNAs that target DNA near the centromere, along with ubiquitous expression of GAL80. As such, nMAGIC can be used to generate clones that are not labelled, whereas pMAGIC (along with a GAL4 line and UAS-marker) can be used much like MARCM to positively label a clone of cells. This manuscript introduces MAGIC transgenic reagents that allow all 4 chromosomes to be targeted. They demonstrate its use in a variety of tissues, including with mutants not compatible with current FLP/FRT methods, and also show it works well in tissues that prove challenging for FLP/FRT mosaic analyses (such as motor neurons). They further demonstrate that it can be used to generate mosaic clones in non-melanogaster hybrid tissues. Overall, this work represents a valuable improvement to the MAGIC method that should promote even more widespread adoption of this powerful genetic technique.

Strengths:

(1) Improves the design of the gRNA-marker by updating the gRNA backbone and also the markers used. GAL80 now includes a DE region that reduces the perdurance of the protein and thus better labeling of pMAGIC clones. The data presented to demonstrate these improvements is rigorous and of high quality.

(2) Introduces a toolkit that now covers all chromosome arms in Drosophila. In addition, the efficiency of 3 target different sites is characterized for each chromosome arm (e.g., 3 different gRNA-Marker combinations), which demonstrate differences in efficiency. This could be useful to titrate how many clones an experimenter might want (e.g., lower efficiency combinations might prove advantageous).

(3) The manuscript is well written and easy to follow. The authors achieved their aims of creating and demonstrating MAGIC reagents suitable for mosaic analysis of any Drosophila chromosome arm.

(4) The MAGIC method is a valuable addition to the Drosophila genetics toolkit, and the new reagents described in this manuscript should allow it to become more widely adopted.

Weaknesses:

(1) The MAGIC method might not be well known to most readers, and the manuscript could have benefited from schematics introducing the technique.

(2) Traditional mosaic analyses using the FLP/FRT system have strongly utilized heat-shock FLPase for inducible temporal control over mitotic clones, as well as a way to titrate how many clones are induced (e.g., shorter heat shocks will induce fewer clones). This has proven highly valuable, especially for developmental studies. A heat-shock Cas9 is available, and it would have been beneficial to determine the efficiency of inducing MAGIC clones using this Cas9 source.

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Author response:

> ***Reviewing Editor Comments:***
>
> *The following are some consolidated review remarks after discussions amongst all three reviewers:*

The reviewers feel the evidence level could be raised from 'convincing' to 'compelling' if the following key (and partially shared) suggestions by the reviewers are followed adequately:

> *(1) Expand labeling options for nMAGIC, which is currently just a BFP marker. This would increase the utility of the method. A far-red marker would be very helpful. Could the authors just do this for one chromosome arm and make the reagent available for others to generate other chromosome arms?*

This is a great suggestion. We will make an nMAGIC vector containing a far-red fluorescent marker and generate a 40D2 version of this nMAGIC gRNA-maker to demonstrate its utility. This vector will be available for others to make additional nMAGIC gRNA-markers.

> *(2) Verify that destabilized GAL80 is potent enough to suppress GAL4. Repeat Figure 1C-E with tub-GAL80-DE-SV40.*

We will use a tub-GAL80-DE-SV40 gRNA-marker to test suppression of pxn-Gal4.

> *(3) Concern about the health of the induced mitotic clones. This is an important consideration, but the reviewers were not sure what the necessary experiments would be. To gauge twin-spot clone sizes? Please address.*

We will assess the health of induced mitotic clones in wing imaginal discs. We will do this by generating twin spots with a nMAGIC gRNA-marker in wing discs and compare the sizes of the two cell populations ($BFP^{+/+}$ and $BFP^{-/-}$) in twin spots.

> *(4) Include a schematic of the MAGIC method as Figure 1 or add it to Figure 1. Many may not be familiar with the method, so to promote its adoption, the authors should clearly introduce the MAGIC method in this paper (and not rely on readers to go to previous publications). For this paper to become a MAGIC reference paper, it should be self-contained.*

We will add a diagram of the MAGIC method in the revised manuscript.

> *(5) Determine the utility of using a hs-Cas9 line for temporal induction of MAGIC clones. This is a traditional method for mitotic clone induction (with hsFLP/FRTs), and its use with the MAGIC system (especially pMAGIC) could also make it more attractive, especially to label small populations of neurons born at known times. To this point, the authors could generate pMAGIC clones using hs-Cas9 for commonly used adult target neurons, such as projection neurons, central complex neurons, or mushroom body neurons. The method to label small numbers of these adult neurons is well worked out with known GAL4 lines, and demonstrating that pMAGIC could have similar results would capture the attention of many not familiar with the pMAGIC method.*

We thank the reviewers for this suggestion. We will test hs-Cas9 in inducing pMAGIC clones in one of the neuronal populations in the adult brain, as suggested by the reviewers.

In addition, we will address all other minor concerns of the reviewers.

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